



Lecture 26

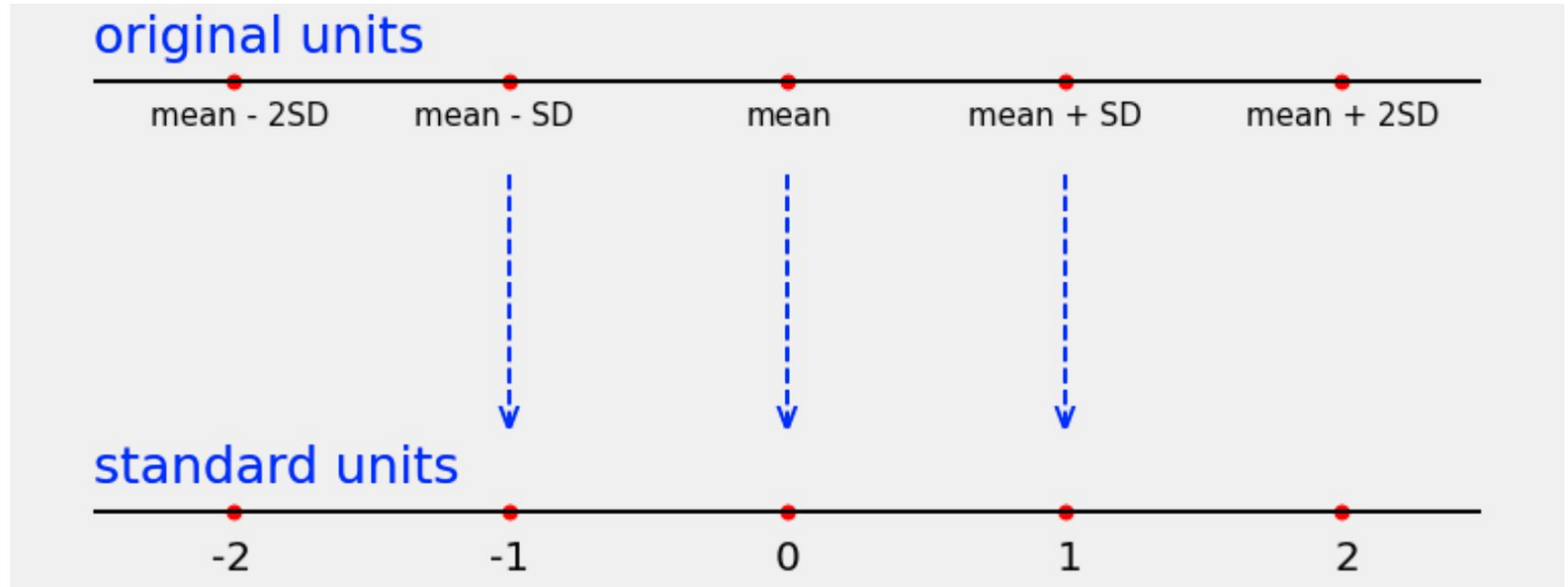
The Normal Distribution

Announcements

- **Homework 8** due Wednesday at 5pm
 - **Lab 8** due Friday at 5pm
 - **Midterm Regrade Requests** are due Friday at 11:59pm
 - **My OH** are today, 3-5pm @ FSM
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Standard Units

Standard Units



$$\text{standard units} = (\text{original value} - \text{mean}) / \text{SD}$$

Standard Units

- Measures: How many SDs above average?
- **$z = (\text{value} - \text{average})/\text{SD}$**
 - Negative z: value below average
 - Positive z: value above average
 - $z = 0$: value equal to average
- When values are in standard units: average = 0, SD = 1

(Demo)

Discussion Question

Find whole numbers
that are close to:

(a) the average age

(b) the SD of the ages

(Demo)

Age in Years	Age in Standard Units
27	-0.0392546
33	0.992496
28	0.132704
23	-0.727088
25	-0.383171
33	0.992496
23	-0.727088
25	-0.383171
30	0.476621
27	-0.0392546

... (1164 rows omitted)

The SD and the Histogram

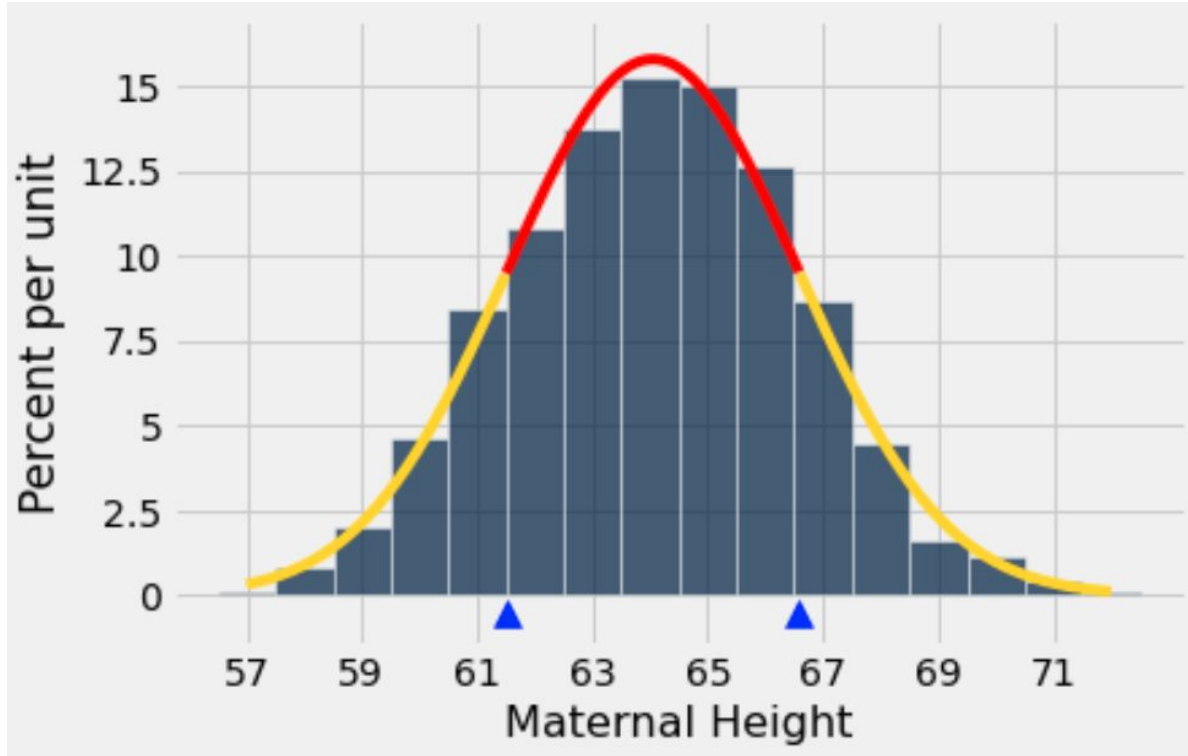
- Usually, it's not easy to estimate the SD by looking at a histogram.
 - But if the histogram has a bell shape, then you can.
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The SD and Bell-Shaped Curves

If a histogram is bell-shaped, then

- the average is at the center
- the SD is the distance between the average and the points of inflection on either side

Points of Inflection



(Demo)

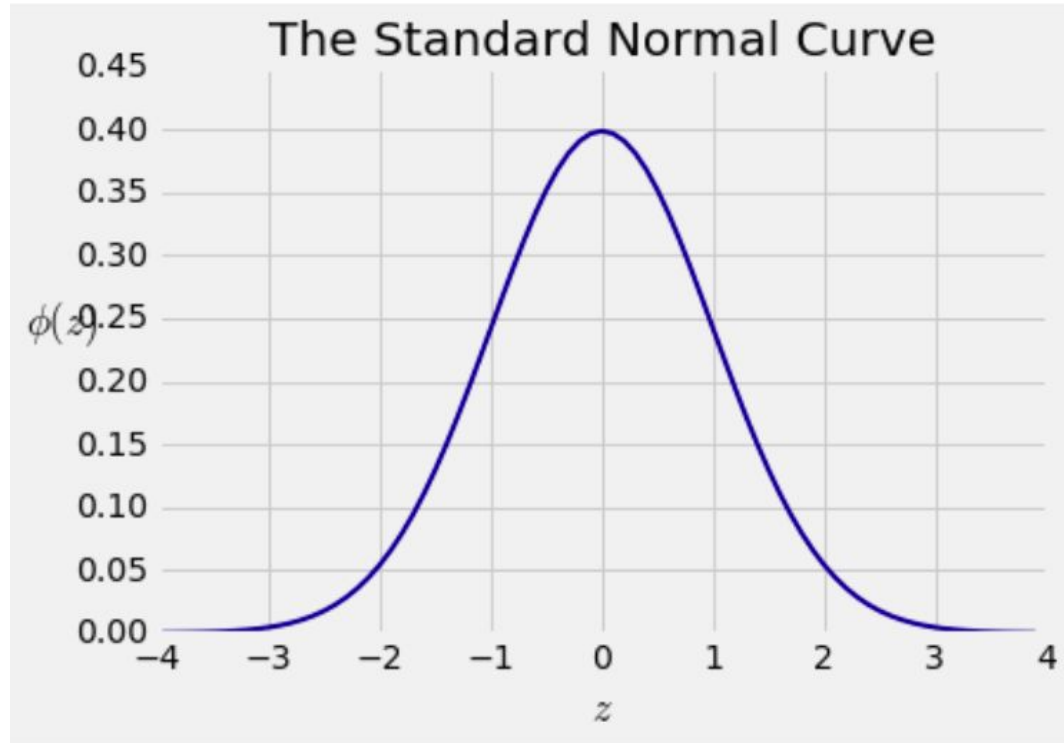
The Normal Distribution

The Standard Normal Curve

A beautiful formula that we won't use at all:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}, \quad -\infty < z < \infty$$

Bell Curve



Normal Proportions

How Big are Most of the Values?

No matter what the shape of the distribution,
the bulk of the data are in the range “average $\pm$ a few SDs”

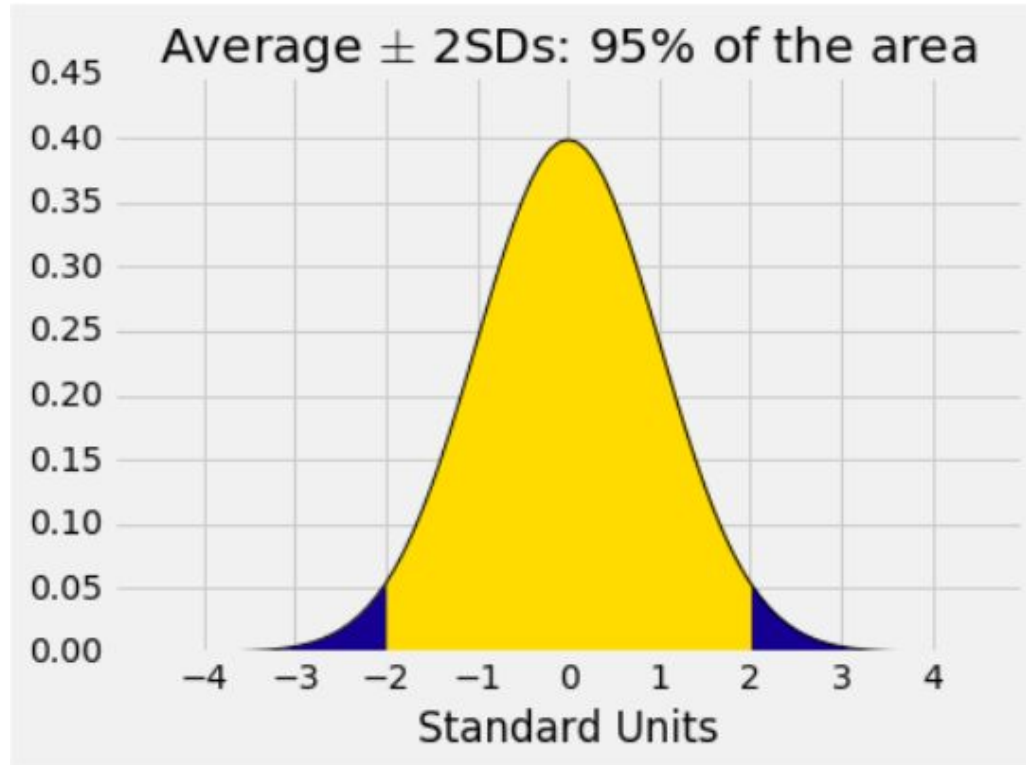
If a histogram is bell-shaped, then

- Almost all of the data are in the range
“average $\pm$ 3 SDs”

Bounds and Normal Approximations

Percent in Range	All Distributions	Normal Distribution
average $\pm$ 1 SD	at least 0%	about 68%
average $\pm$ 2 SDs	at least 75%	about 95%
average $\pm$ 3 SDs	at least 88.888...%	about 99.73%

A “Central” Area



Central Limit Theorem

Sample Averages

- The Central Limit Theorem describes how the normal distribution (a bell-shaped curve) is connected to random sample averages.
- We care about sample averages because they estimate population averages.

(Demo)

Central Limit Theorem

If the sample is

- large, and
- drawn at random with replacement,

Then, *regardless of the distribution of the population,*

the probability distribution of the sample average
is roughly normal

Discussion Question

After rolling 1,000,000 fair 6-sided dice, which of these histograms would you expect to have a bell shape? Check all that apply.

- 1) The histogram of outcomes of these million rolls
 - 2) The histogram that results from computing the average outcome of these million rolls
 - 3) The histogram that results from splitting the outcomes into 1,000 groups of 1,000 (in the order they occurred) and computing the average outcome of each group
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